Mining Dairy Data with AI Agents

Pádraig Lucey

What is an agent?

[An automation that] runs tools in a loop to achieve a goal.

Simon Willison

'AI' is not a tool.

Rich Hickey



Goal

Tell me about pen 21's intake over the last month, what's changed, and how the cows are responding.

Worked for 4 s · 3 steps | New corn silage pile | ½ lb more bypass fat | ½ lb less rolled corn

✓ Opening pen 21 `open_pen(21)` 75 days

Pen 21

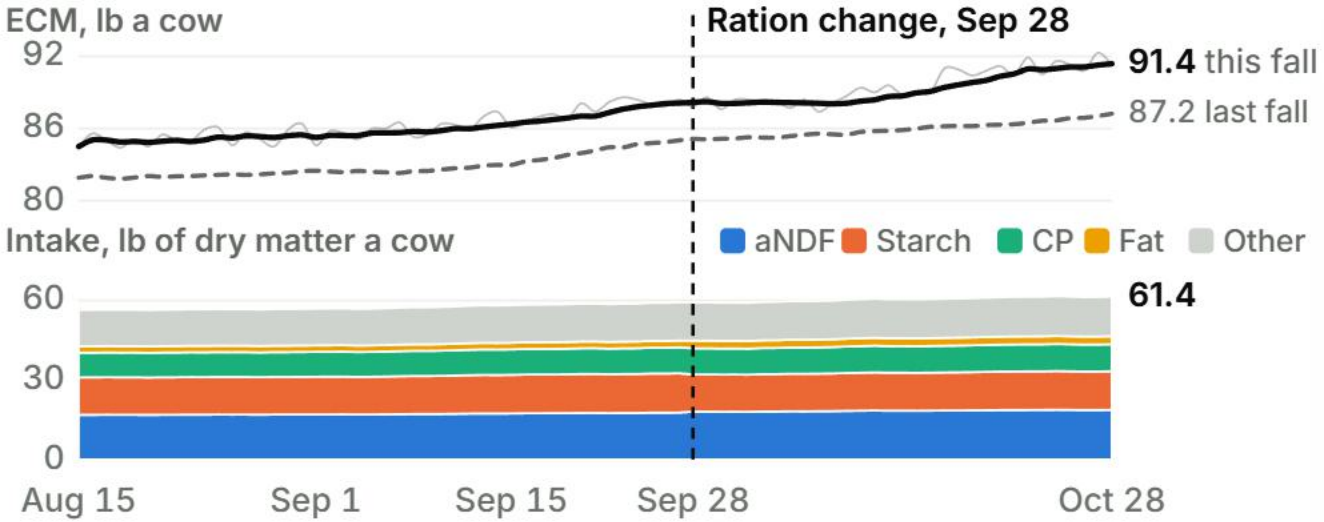
238 cows · High cows · Aug 15 to Oct 28

Milk a cow, last test
90.6 lb Highest of 12 pens
Middle 80%: 73–108 lb

Days in milk
52 median
Middle 80%: 31–74 days

Lactation
3rd median
2nd 84 · 3rd 51 · 4th+ 103

Intake, last 7 days
61.4 lb formulated 63.0
Daily, the last 30 days



Maybe this change is just seasonal?

Checking the same weeks last fall.

ECM is up 3.3 lb since Sep 28. Last fall, same weeks, no ration change

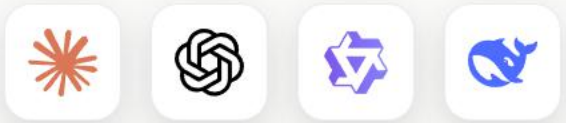
Goal

Tell me about pen 21's intake over the last month, what's changed, and how the cows are responding.

Model, harness, agent

The model

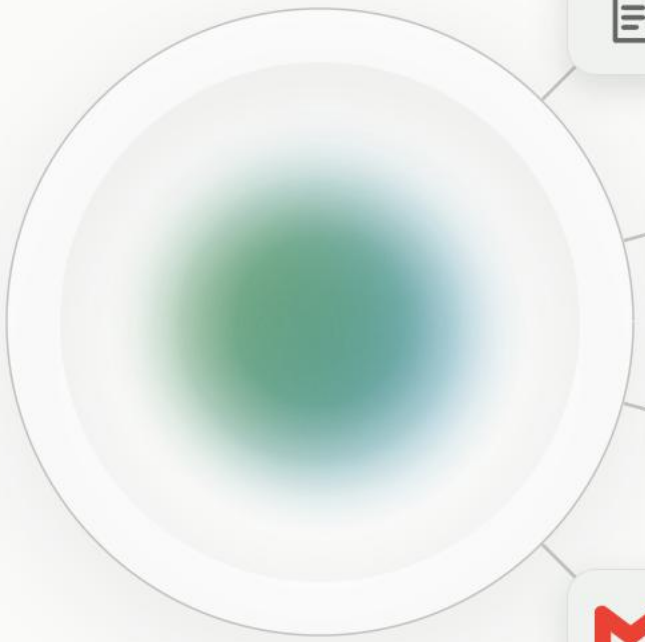
Reads your prompt and predicts the words that come next.



The harness

Gives the model its tools, and runs them.





Tools

Read a data file
↩ Cow file and feed report

Weather station API
↩ Temperatures, both weeks

Run statistics
↩ A difference from Sep 28

Email MCP
↩ 2 lab reports

After Harness Engineering 101, Isuru Wijesiri, 2026

AGENTS

Nutrition · Herd A · close-up pen

No Close Up Mineral in the 13 Oct premix

Close Up Mineral	**0** of 1,410 lb
Fed to the close-up pen	**14–16** Oct
The 12 batches before	1,450–2,110 lb



HOW WE KNOW

The wagon scale gained 0 lb during the Close Up Mineral step. EZfeed loading records.

Close-up minerals often carry the anionic salts that lower milk fever risk.
Santos et al., J Dairy Sci 2019

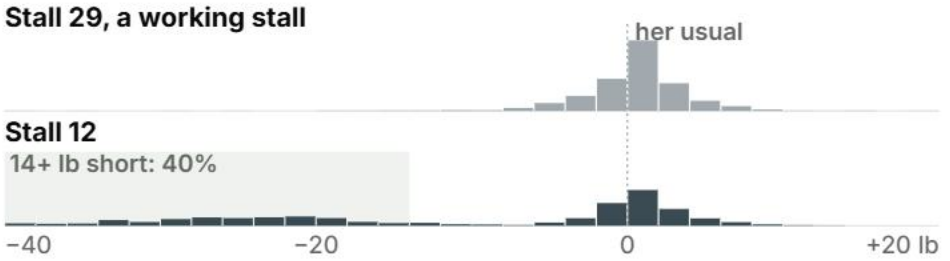
Did the 13 Oct batch get its Close Up Mineral?

Yes | No | Don't know

Milk · Herd B · rotary parlor

3 stalls milk low

Stall 12	**−10.2** lb a milking
Stall 57	**−8.9** lb
Stall 73	**−7.8** lb
The other 77 stalls	−2.9 to +2.4 lb



HOW WE KNOW

40% of stall 12's milkings came in 14 lb or more short of the cow's usual; 1% in a working stall.

Meter or milking unit? A meter reading short loses nothing; a unit not milking out leaves about 1,300 lb a day in the cows.
Andreen et al., J Dairy Sci 2020

Have stalls 12, 57 and 73 been checked?

Yes | Not yet | Don't know

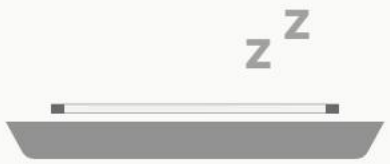
HARNESS

Where it runs

5:00 am

HARNESS Installed on the machine or In a sandbox

MODEL On the machine, like the Rossow lab or A frontier model, through its API



Your laptop

Only while it's open

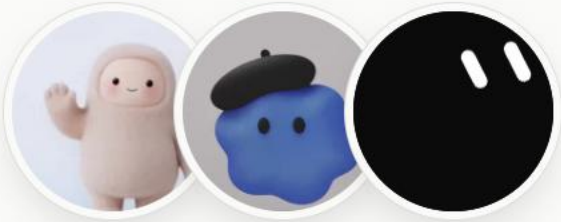
○ Lid shut, asleep



A box in the office

A Mac mini, or a $5 Ubuntu server

● Running



Their cloud

Muse, dots, Grok Bot

● Running

FOUND AT 5:00

NUTRITION New haylage sample added to your feed library

MILK Bulk tank SCC up from 180,000 to 260,000 this week

TOOLS

Dairy Data Tools

```
$ claude plugin marketplace add padge01/dairy-data-tools
```

```
$ pip install dairy-data-tools
```

Pen feed

3 milking pens · 7 days to Oct 28

Pen 1

15 cows

ECM per lb DM	DM fed	Feed per cwt
1.62	**44.6** lb	**$10.58**

FORAGES	
Corn silage, 2025 pile	**13.8** lb
Alfalfa hay	**9.8** lb
MINERALS AND ADDITIVES	
Lactation mineral	**1.2** lb
BY-PRODUCTS	
Almond hulls	**1.5** lb
PROTEINS	
Canola meal	**4.9** lb
Soybean meal	**2.9** lb
GRAINS AND ENERGY	
Rolled corn	**8.9** lb

Pen 11

82 cows

ECM per lb DM	DM fed	Feed per cwt
1.20	**46.3** lb	**$14.15**

FORAGES	
Corn silage, 2025 pile	**12.1** lb
Alfalfa hay	**6.4** lb
Wheat silage	**1.9** lb
MINERALS AND ADDITIVES	
Lactation mineral	**1.1** lb
FATS	
Bypass fat	**0.4** lb
BY-PRODUCTS	
Whole cottonseed	**3.0** lb
Almond hulls	**2.6** lb
PROTEINS	
Canola meal	**4.5** lb
Soybean meal	**1.9** lb
Distillers grains	**1.5** lb
GRAINS AND ENERGY	
Rolled corn	**9.4** lb
Molasses	**0.4** lb

Pen 21

129 cows

ECM per lb DM	DM fed	Feed per cwt
1.43	**63.0** lb	**$12.34**

FORAGES	
Corn silage, 2026 pile	**17.0** lb
Alfalfa hay	**8.7** lb
Wheat silage	**2.4** lb
MINERALS AND ADDITIVES	
Lactation mineral	**1.5** lb
FATS	
Bypass fat	**1.0** lb
BY-PRODUCTS	
Whole cottonseed	**3.9** lb
Almond hulls	**3.4** lb
PROTEINS	
Canola meal	**6.8** lb
Soybean meal	**2.4** lb
Distillers grains	**1.9** lb
GRAINS AND ENERGY	
Rolled corn	**11.7** lb
Molasses	**0.5** lb

Cow 3256

Pregnant 147 d

Pen 21 · Holstein · 5th lact

77 lb, last test — **245** DIM

Bottom 40% of pen 21, last test (129 cows)

Her milk against pen 21, last test (lb)



Lowest 16 · her 77 · highest 162

Her milk this lactation, lb by days in milk (each test)



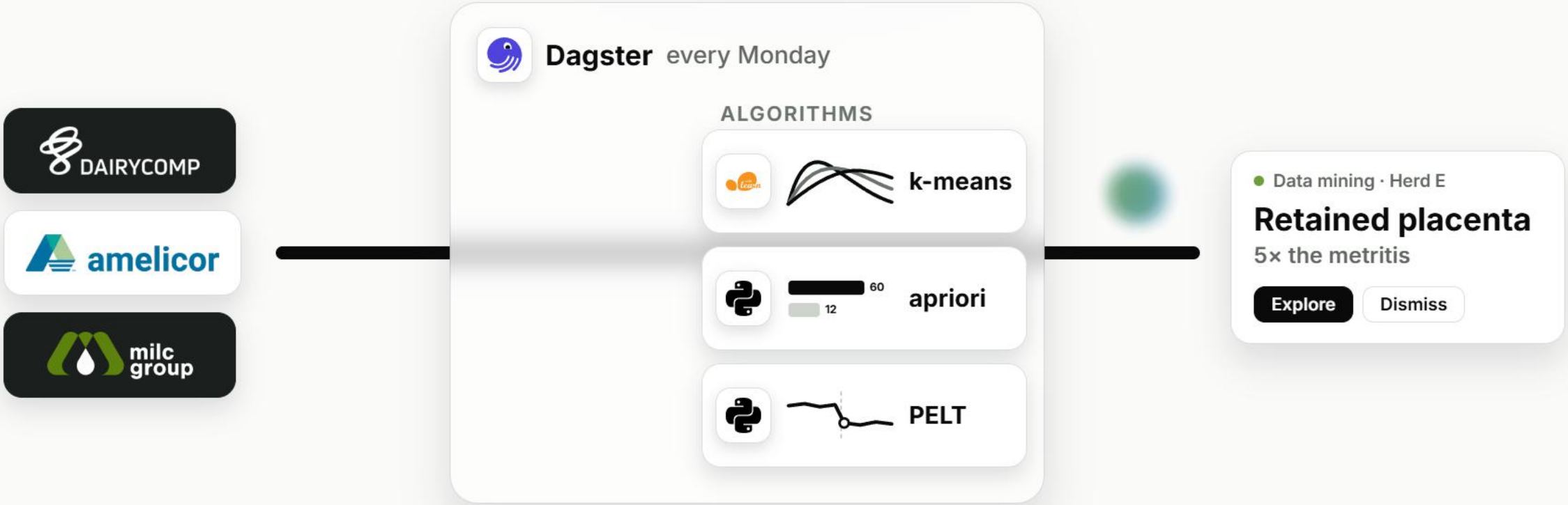
Fat · protein	ECM, last test	SCC
3.9% · 3.6%	**85 lb**	**138k**

Jul 30	**Footbath**
Jul 29	**Lameness**
Jul 6	**Checked pregnant**
Jun 3	**Bred**

+4 earlier this lactation

DATA MINING

Agent-run ML pipelines



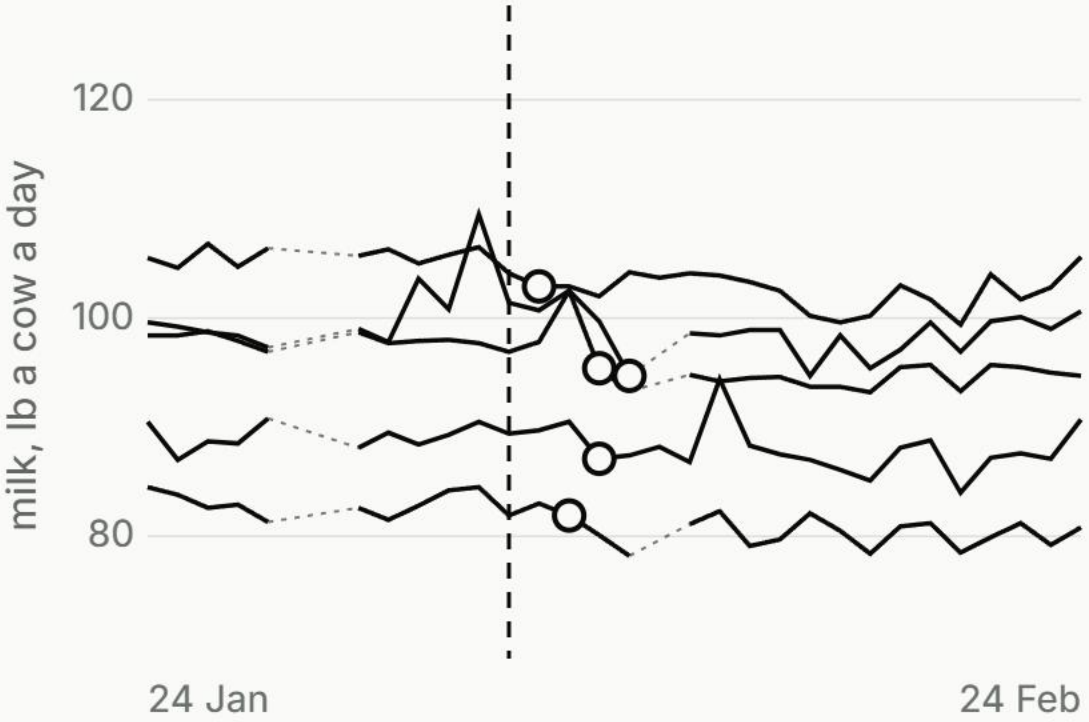
DATA MINING

Feed switches, milk response

Herd A: 39 of the 59 shifts came after a ration change in that pen

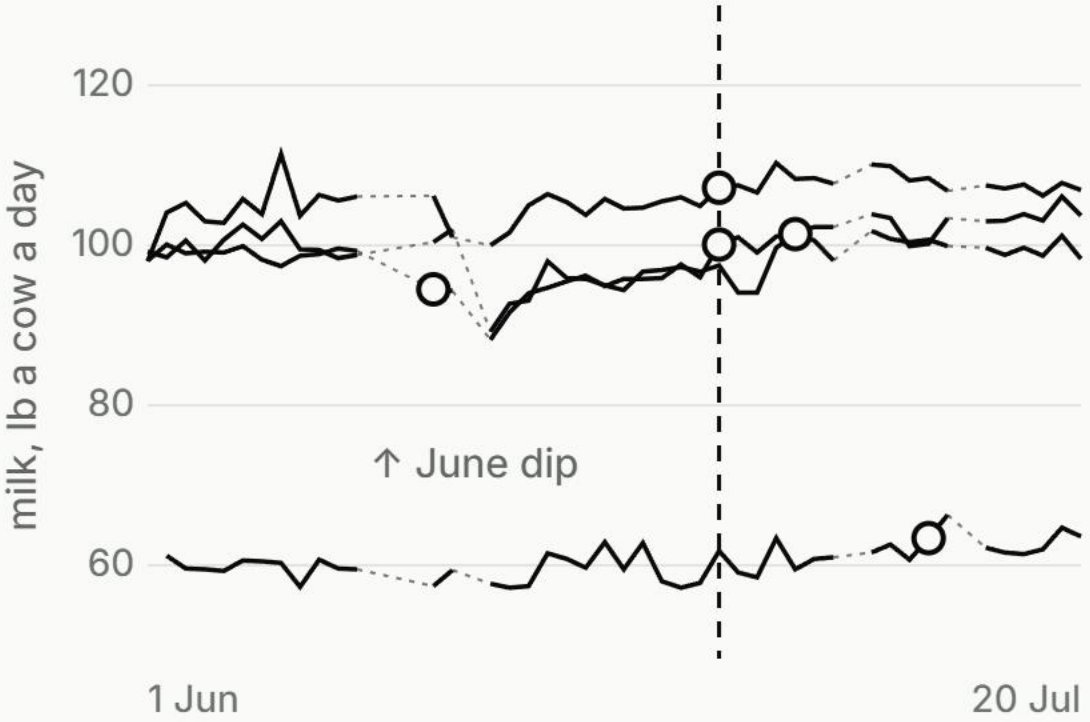
Sorghum for wheat silage, 5–10 Feb

5 pens: −2 to −4 lb



Dry distillers for wet, 1–3 Jul

4 pens: +3 to +7 lb

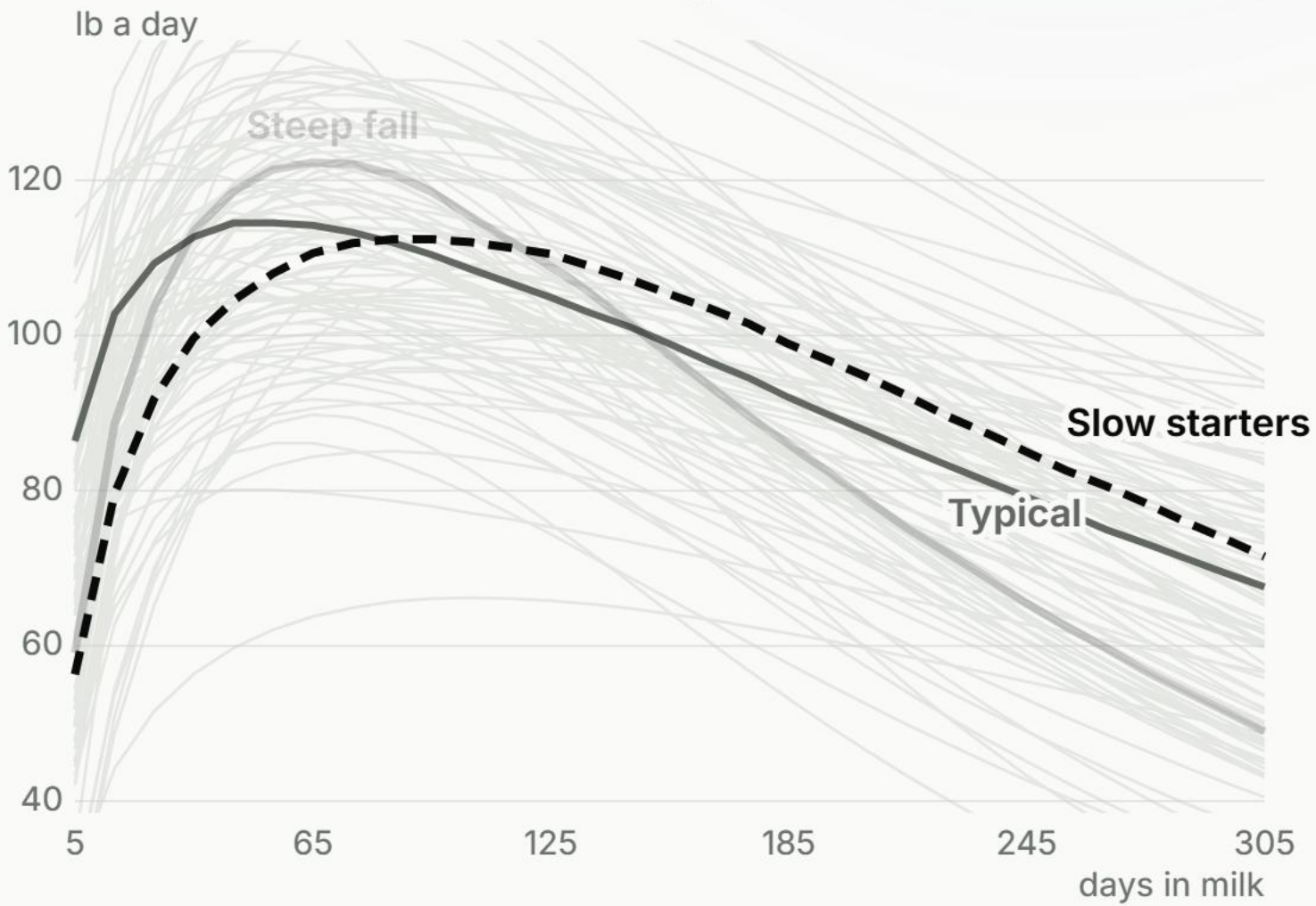


`PELT change points · Herd A`

DATA MINING

Three shapes of lactation

The median curve of each shape



MASTITIS, FIRST 30 DAYS · HERD A

Slow starters
396 cows

6.8 in 100

Typical
1,037 cows

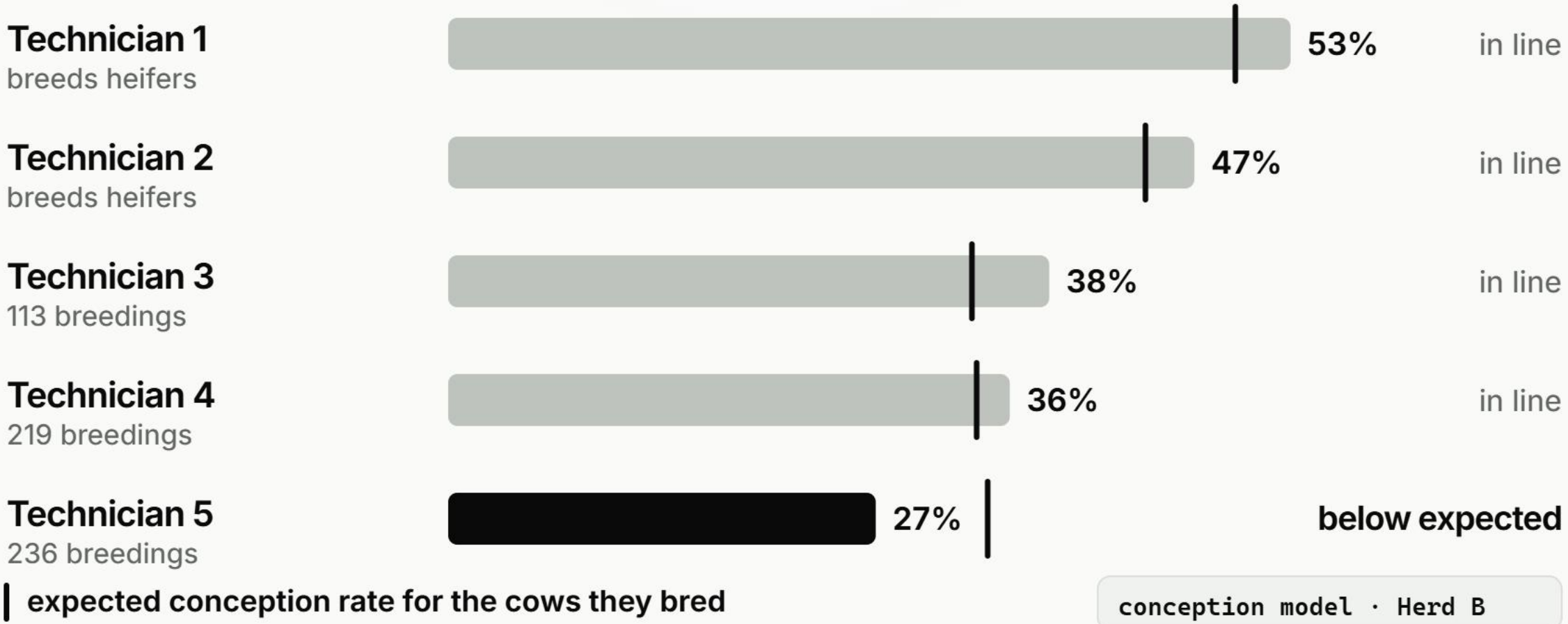
3.1 in 100

`k-means, 4 groups → 3 shapes`
`2,706 lactations · 6 dairies`

DATA MINING

Technicians, on the cows they bred

Herd B · pregnant per breeding



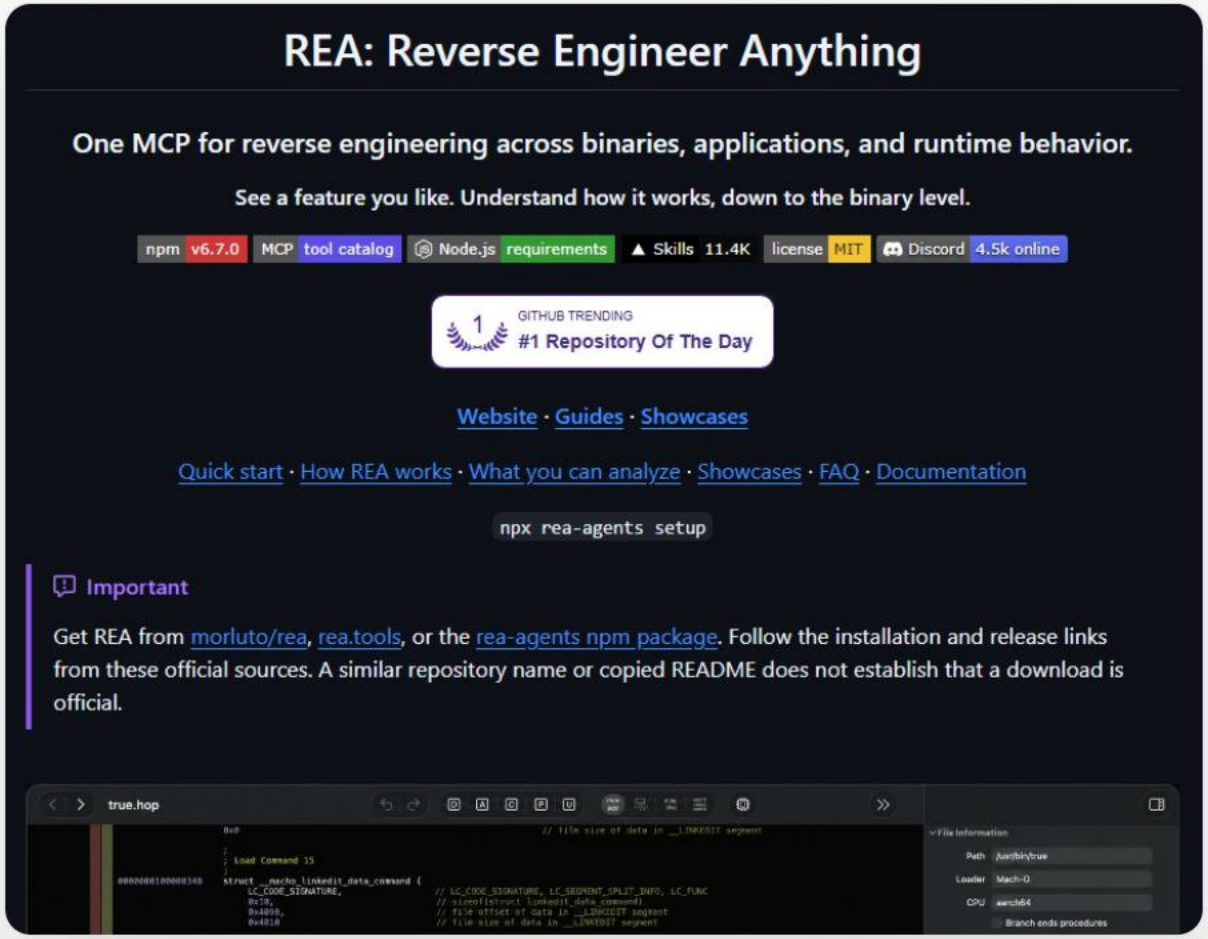
conception model · Herd B

REVERSE ENGINEERING

Reverse engineer anything

`github.com/morluto/rea`

One install line gives an agent the tools to take software and machines apart.



github.com/morluto/rea, 27 Oct 2026

★ 397,785

stars in six months

99,759 copies people took to build on

- Old hardware
- Smart TVs
- Apps
- Farm machines

REVERSE ENGINEERING

The calf milk pasteurizer

THE NEED

Its alerts only reach a phone standing next to it

The machine talks Bluetooth, to its own phone app

HOW WE READ IT

A Raspberry Pi listens for the same Bluetooth signal

The temperature and minutes left, decoded from the bytes, every few seconds

WHAT IT DOES

Works out each cycle and texts the crew when bags are ready

In English or Spanish. Controls cycles remotely.

SINCE 31 JULY

227 cycles and 1,833 texts at two dairies

The machine is untouched





Thank you

Pádraig Lucey MVB, MPVM, Ph.D., DACVPM-Epi, PAS
Dairy Technical Consultant, Elanco Animal Health
padraig.lucey@elancoah.com

Megan Mouw B.S.
Territory Account Manager
megan.mouw@elancoah.com

Clay Daily B.S.
Channel Representative
clay.daily@elancoah.com

